

OncoEvidence: Mechanism-Guided AI Evidence Triage for Cancer Drug Repurposing

Mechanism-grounded AI triage that beats keyword evidence screening on precision and matches expert manual curation at scale

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The problem. Drug repurposing seeks new uses for existing drugs, for example whether an approved drug could treat a cancer it was not designed for. Models can rank thousands of drug-disease pairs, but a score alone is not actionable: a reviewer needs the *mechanism* (why the drug might work) and *evidence* (whether the literature supports it). OncoEvidence ranks oncology candidates, extracts the mechanism path that would explain each one, and uses a language model to verify that mechanism against published abstracts.

Thesis. The contribution is a *methodology comparison*, not a single model. *Mechanism-grounded AI triage* (knowledge-graph mechanism extraction plus a literature-grounded language model) is measured against the two status-quo ways to vet repurposing candidates: a manual / expert-curated process, and naive automated screening.

- **Beats keyword screening.** The language-model verifier identifies real mechanisms with **0.857 precision**, versus **0.591** for keyword / co-occurrence screening of the same abstracts.
- **Matches expert manual curation, automatically.** The extracted mechanisms agree with the expert-curated DrugMechDB at **0.802**, reproducing by pipeline what is normally hand-curated, across hundreds of pairs.
- **Honest baseline check.** On the plain link-ranking task a tuned tabular model (XGBoost) matches or beats the graph network, so the graph earns its place by producing the traceable *mechanism* a tabular model cannot, not by the score. Mechanism paths separate true from random indications at **AUROC 0.879**.

1 Background and Significance

Developing a new drug takes many years and large cost, while thousands of approved drugs already have known safety profiles. If an approved drug acts on a protein or pathway that also drives a cancer, it can be a fast route to a new treatment. Biomedical knowledge graphs such as PrimeKG encode millions of relationships among drugs, proteins, genes, pathways, and diseases, so the information needed to reason about such mechanisms exists as paths through the graph. The open problem is *trust*: a bare drug-disease score hides its reasoning, and cannot separate a real mechanism from a coincidence. OncoEvidence addresses this by making the mechanism explicit and checking it against the literature and a curated database.

2 Central Question and Specific Aims

Core question. For cancer drug repurposing, can a biomedical knowledge graph combined with a literature-grounded language-model agent prioritize candidates by both predicted therapeutic relevance and mechanistic plausibility, and when does the graph add value over a strong tabular baseline?

Specific Aims

Aim 1 (Candidate generation). Rank drug-cancer pairs with a heterogeneous graph network and a knowledge-graph embedding, against a tuned XGBoost baseline on identical features; report honestly where the graph helps.

Aim 2 (Mechanism extraction). Extract multi-hop *drug* $\rightarrow$ *target* $\rightarrow$ *pathway/cancer-gene* $\rightarrow$ *cancer* paths, down-weighting promiscuous hubs.

Aim 3 (Evidence verification). A language model grades each path supported / weak / contradicted / unknown over retrieved abstracts, with a verbatim quote.

Aim 4 (Evaluation). Separation of true vs random pairs, agreement with curated Drug-MechDB mechanisms, and verifier precision.

3 Datasets and System

Resource	Content	Role
PrimeKG	129,312 nodes, 8.1M edges, 50 relation types; 7,957 drugs, 17,080 diseases (3,170 oncology), 9,388 indication edges	Ranking + mechanism paths
Europe PMC	abstracts and open-access full text via REST	Evidence retrieval
DrugMechDB	curated drug mechanism-of-action paths (UniProt-keyed)	Mechanism ground truth

Table 1: Public resources used. PrimeKG supplies the graph; Europe PMC the evidence; Drug-MechDB the curated mechanisms (mapped UniProt $\rightarrow$ HGNC via mygene.info).

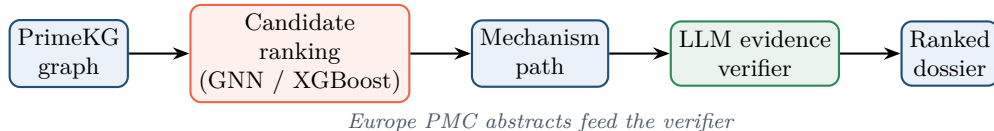


Figure 1: The OncoEvidence pipeline: rank candidates, extract a mechanism path, verify it against the literature, output an evidence-backed shortlist.

4 Methods

4.1 Candidate ranking and the honest baseline

Repurposing is framed as link prediction on *drug* $\rightarrow$ *indication* $\rightarrow$ *disease* edges. A heterogeneous graph neural network encodes each node by message passing over its typed neighbours; with a

GraphSAGE-style layer,

$$h_v^{(l)} = \sigma\left(W^{(l)} \left[h_v^{(l-1)} \parallel \text{AGG}_{u \in \mathcal{N}(v)} h_u^{(l-1)} \right]\right), \quad s(d, c) = \sigma(z_d^\top R z_c), \quad (1)$$

where $s(d, c)$ scores a drug-disease pair. The graph network, a DistMult knowledge-graph embedding, and a tuned XGBoost baseline all receive the *same* SentenceTransformer node features, so any gap measures the value of graph structure, not features.

4.2 Mechanism-path extraction

For a candidate we search PrimeKG for short paths through mechanism relations (drug-target, protein-protein, protein-pathway, protein-disease), scoring each by specificity and penalizing promiscuous hubs:

$$\text{spec}(x) = \frac{1}{\log_2(\deg(x) + 2)}, \quad \text{score}(p) = \beta_{\text{type}} + \text{spec}(\text{bridge}) \cdot \frac{1}{\log_2(\delta_{\text{drug}} + 2)}, \quad (2)$$

where δ_{drug} is the number of drugs that bind the bridge protein (an inverse-frequency penalty that suppresses generic carriers such as albumin), and β_{type} orders direct-target (3) over interaction (2) over shared-pathway (1) templates.

4.3 Evidence verification

For the top path we retrieve abstracts (a multi-query Europe PMC merge biased toward mechanism statements), extract the sentences mentioning both the drug and the target gene, and ask a language model to grade the mechanism as supported / weak / contradicted / unknown, requiring a verbatim quote and treating a purely statistical association as *weak*, not supporting.

4.4 Evaluation metrics

We report AUROC for separation, agreement with curated mechanisms, and the precision of “supported” verdicts:

$$\text{precision} = \frac{\#\{\text{“supported” pairs whose path gene is in the DrugMechDB MOA set}\}}{\#\{\text{“supported” pairs the DrugMechDB covers}\}}. \quad (3)$$

5 Preliminary Results

5.1 Finding 1: the link task does not need the graph

With a properly tuned XGBoost on shared features, the graph network does not win in any regime (Table 2). The earlier impression that it did was traced to an untuned baseline and a message-passing leak, both fixed. We report this plainly: for link ranking with rich text features, a strong tabular model suffices.

Test AUROC (5 seeds)	GNN	XGBoost	MLP	KGE
Transductive	0.977	0.988	0.973	0.907
Inductive (cold-disease, oncology)	0.882	0.963	0.930	0.452
Inductive (cold-drug)	0.956	0.958	0.931	0.482

Table 2: Corrected benchmark. A tuned XGBoost matches or beats the GNN everywhere; the DistMult memorizer (KGE) collapses on unseen nodes, confirming the task needs content features, not memorized identity.

5.2 Finding 2: the graph carries real mechanism

A tabular model cannot produce a traceable mechanism; the graph can. On true oncology indications the extractor recovers textbook direct-target mechanisms:

```
Quizartinib --targets--> FLT3 <--associated-- myeloid leukemia
Tamibarotene --targets--> RARA <--associated-- acute promyelocytic leukemia
Topotecan --targets--> TOP1 <--associated-- ovarian cancer
Trifluridine --targets--> TYMS <--associated-- colorectal cancer
```

Over 400 true indications vs 400 random drug-cancer pairs, the mechanism signal separates the two groups strongly (Table 3, Fig. 2).

Metric	True	Random
Mean mechanism score	1.95	0.17
Direct-target rate	34.0%	0.25%
Any mechanistic path	81.3%	8.8%

Separation AUROC
0.879

Table 3: Mechanism paths separate true indications from random pairs. A true indication is roughly 136× more likely to have a direct drug-target to cancer-gene link than a random pair.

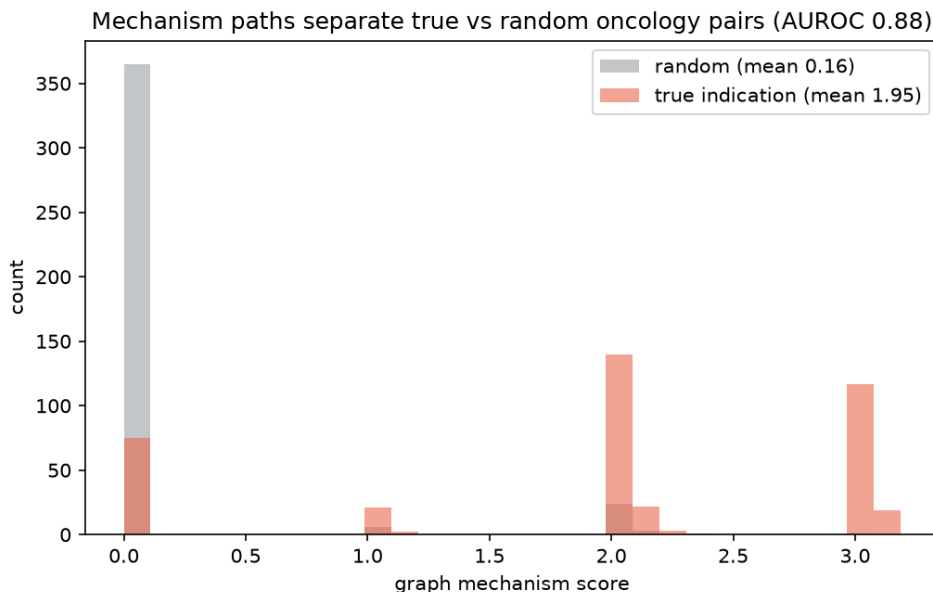


Figure 2: Distribution of the graph mechanism score for true indications (orange) vs random drug-cancer pairs (grey). Random pairs pile up at zero; true indications spread to high, specific scores (AUROC 0.879).

5.3 Finding 3: the mechanisms agree with a curated database

Mapping DrugMechDB’s UniProt accessions to HGNC symbols makes a direct comparison possible. On the 263 true pairs DrugMechDB covers, our extracted bridge genes overlap the curated mechanism genes for **211 pairs (agreement 0.802)**.

Drug	Overlapping curated genes	
Methotrexate	ATIC, DHFR, TYMS	✓
Topotecan	TOP1	✓
Decitabine	DNMT1, DNMT3A	✓

Table 4: Examples of agreement with curated DrugMechDB mechanisms. DHFR is methotrexate’s canonical target, recovered by the extractor.

5.4 Finding 4: the language model verifies precisely

Run over 50 true vs 50 random pairs, the verifier marks far fewer “supported” than a keyword baseline, but those calls are much more trustworthy (Table 5). A stricter rubric (an explicit drug-to-target statement is required; a genetic or statistical association is graded weak) plus sentence-level grounding collapsed “unknown” from 14 to 1.

	Language model	Keyword baseline
True pairs graded “supported”	11 / 50	34 / 50
Random pairs graded “supported”	1 / 50	2 / 50
Precision of “supported” vs DrugMechDB	0.857 (6/7)	0.591 (13/22)

Table 5: The language model is the conservative, precise reviewer: of its “supported” calls that DrugMechDB covers, 86% are confirmed, against 59% for keyword matching. Each verdict carries a verbatim quote, e.g. Topotecan / TOP1: “*a topoisomerase I (TOP1) inhibitor*”.

6 Discussion: what the method beats (and what it does not)

The contribution is best read as a comparison of three ways to vet cancer repurposing candidates. Against *keyword / co-occurrence screening*, the language-model verifier is far more precise (0.857 vs 0.591), because it reads whether the text actually states a mechanism rather than whether the words merely co-occur (a keyword screen grades a pharmacogenetic association as “supported”; the language model correctly downgrades it to weak). Against the *manual curation* process, the pipeline reproduces expert DrugMechDB mechanisms at 0.802 automatically and at scale, doing in seconds per candidate what manual literature review does by hand. We are explicit about what the method does *not* beat: on the plain link-ranking task a tuned tabular model is as good as the graph, so we do not claim a model win there; the graph earns its place by producing the traceable mechanism that a tabular model cannot.

Limitations, stated plainly. We do not claim to beat human experts on *quality*; we match curated quality while automating and scaling it. The separation partly reflects that drugs indicated for a cancer already have a known target linked to that cancer in the graph, so this validates that the graph encodes *known* mechanism consistently, rather than discovering new biology. Some metrics rest on modest samples (the verifier precision on seven covered “supported” pairs) and partial coverage (DrugMechDB; the open-access full-text subset). The planned summer work scales the verifier run, broadens full-text coverage, and applies the same triage to *novel* candidate pairs to generate mechanism-verified repurposing hypotheses.

7 Eight-Week Timeline

Weeks	Milestones
1–2	PrimeKG pipeline; leakage-safe splits; shared features.
3–4	Candidate ranking + tuned XGBoost baseline (Finding 1).
5	Mechanism extraction with hub down-weighting (Findings 2–3).
6	Literature retrieval and the verifier (Finding 4).
7	Full evaluation; scale the verifier; error analysis.
8	Deliverable shortlist, figures, written report.

Table 6: Weeks 1–6 are already de-risked by the preliminary results above.

8 Reproducibility and Ethics

Open tools (PyTorch Geometric, scikit-learn, XGBoost) and public data (PrimeKG, Europe PMC, DrugMechDB); code, configs, seeds, and the UniProt→HGNC cache are released, and every reported number is multi-seed where applicable. All predictions are research-only and hypothesis-generating, not medical advice.

References

- [1] Huang K, Chandak P, et al. *A foundation model for clinician-centered drug repurposing (TxGNN)*. Nature Medicine, 2024.
- [2] Chandak P, Huang K, Zitnik M. *Building a knowledge graph to enable precision medicine (PrimeKG)*. Scientific Data, 2023.
- [3] *KGML-xDTD: a knowledge graph-based framework for drug treatment prediction and mechanism description*. 2023.
- [4] Zitnik M, Agrawal M, Leskovec J. *Modeling polypharmacy side effects with graph convolutional networks (Decagon)*. Bioinformatics, 2018.
- [5] Mayers M, et al. *DrugMechDB: a curated database of drug mechanism paths*. 2022.
- [6] Hamilton W, Ying R, Leskovec J. *Inductive representation learning on large graphs (GraphSAGE)*. NeurIPS, 2017.